NJIT DS677: Deep Learning	**Fall 2025**

Homework 0

Due Date: **Mon., Sept 22nd 2025, 11:29 am** *Your Name:*

Instructions: There are **10 problems** in total in this homework. The breakdown of individual scores per sub-problem are provided in this pdf. You may use the provided LaTeX template to typeset your report if you are comfortable with it. Otherwise prepare your submission in a clear, legible manner. Provide sufficient explanations in all solutions but avoid proving lecture or out-of-scope material (unless explicitly asked to).

What to submit: Submit a pdf file containing your answers **online through Canvas** by the due date/time.

Policies: Collaborative reports are not allowed. Even if you discuss problems with classmates, you are expected to write and submit **individual reports**.

Mathematical Background

Probability

Problem 1 [5 points] Consider a random variable X for which the expectation exists and is finite, i.e. $\mathbb{E}[X] < \infty$. We can define the variance of the random variable as $\text{var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$. Prove that $\text{var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Problem 2 [5 points] Consider a random variable x that has the following probability density:

$$p(x) = \frac{1}{\sqrt{8\pi}} e^{-\frac{(x-5)^2}{8}}$$

Evaluate the following integral:

$$\int_{-\infty}^{\infty} (ax^2 - bx + c) p(x) dx$$

Problem 3 [Bonus, 20 points] Suppose we pick a permutation π uniformly at random from the set of all permutations on $[n] := \{1, 2, \ldots, n\}$. Let X be the number of fixed points of π (i.e., the number of $i \in [n]$ such that $\pi(i) = i$).

(a) What is $\mathbb{E}[X]$?

(b) What is $\text{Var}[X]$?

Linear Algebra

In deep learning we will be dealing with data and parameters in high-dimensional spaces. Norms help us compute lengths and distances in those high-dimensional spaces. A typical ℓ_p norm (for $p >= 1$) is given by:

$$||\mathbf{x}||_p = \left(\sum_{i=1}^{d} |x_i|^p \right)^{\frac{1}{p}}$$

For $p = \infty$, the norm is typically defined as $||\mathbf{x}||_\infty = \max_i |x_i|$

Problem 4 [5 points] Figure 1 shows three different norm balls corresponding to the curves $||\mathbf{x}||_2 = 1$, $||\mathbf{x}||_1 = 1$, and $||\mathbf{x}||_\infty = 1$. Match the curves with the equations. The orange curve is labeled **a**, the blue curve is labeled **b** and the magenta curve is labeled **c**.

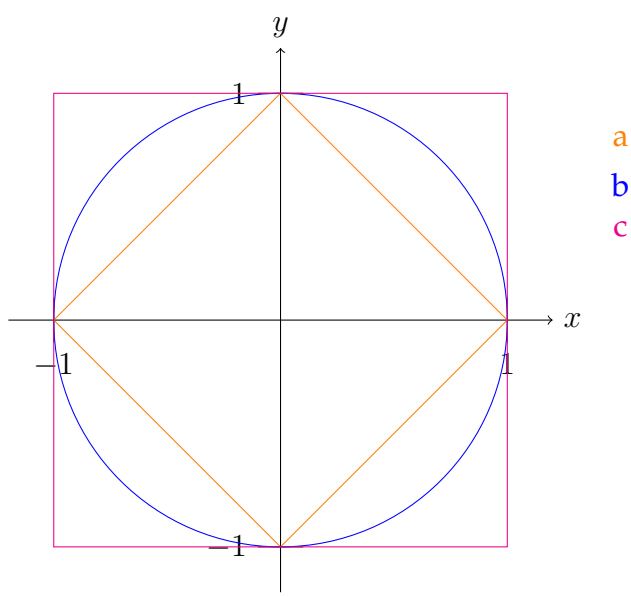

Figure 1. Problem 3

Problem 5 [10 points] As is evident from the previous question, ℓ_p norms with different values of p appear different. However they are still broadly equivalent. Show the following equivalences:

(a) $||\mathbf{x}||_2 \leq ||\mathbf{x}||_1 \leq \sqrt{d}||\mathbf{x}||_2$

(b) $||\mathbf{x}||_\infty \leq ||\mathbf{x}||_2 \leq \sqrt{d}||\mathbf{x}||_\infty$

Hint: You may find the Cauchy-Schwartz inequality helpful in showing at least one of the above.

Problem 6 [10 points] Consider a matrix $\mathbf{X} \in \mathbb{R}^{n\times d}, n \leq d$ with singular values $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_n$. The Frobenius norm $||\mathbf{X}||_F$ is given by $||\mathbf{X}||_F = \sum_{i,j=1,1}^{n,d} X_{ij}^2 = \sqrt{\text{trace}(\mathbf{X}^\top\mathbf{X})}$.

(a) Using this definition prove that $||\mathbf{X}||_F = \sqrt{\sum_{i=1}^{n} \sigma_i^2}$. You may use the singular value decomposition $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ where $\mathbf{U} \in \mathbb{R}^{n\times n}$ and $\mathbf{V} \in \mathbb{R}^{d\times d}$ are orthonormal matrices, and $\mathbf{\Sigma}$ is a rectangular matrix with non-zero entries only on the main diagonal.

(b) Prove that $||\mathbf{X}||_F \leq \sqrt{n}||\mathbf{X}||_2$ where $||\mathbf{X}||_2 = \sigma_1$ is the spectral norm of $\mathbf{X}$

Calculus

Problem 7 [5 points] Consider the following scalar-valued function of a vector, called the log-sum-exp function. It is defined as:

$$f : \mathbb{R}^d \to \mathbb{R}, f(\mathbf{x}) = \log\left(\sum_{i=1}^{d} e^{x_i}\right)$$

Find the gradient of f wrt $\mathbf{x}$. Can you connect this to an operation you might have seen in a machine learning class?

Problem 8 [10 points] Show that $f(\mathbf{x})$ can be viewed as an approximation of the $\max$ function. Specifically, show the following bounds:

$$\max\{x_1, \ldots, x_d\} \leq f(\mathbf{x}) \leq \max\{x_1, \ldots, x_d\} + \log(d)$$

Note that this also makes the transformation represented by the gradient of f, an approximate indicator vector for the maximum entry of $\mathbf{x}$.

Pytorch Warmup

Problem 9 [10 points] Complete the following exercises. Include the lines of code in the same pdf as your other answers. Do not attach outputs of executed code. These exercises are common steps in the language modeling pipeline.

1. Create two random tensors with gaussian entries of size `(29, 30, 32)` and `(32, 100)`. This is how we produce embeddings for a batch of sentences. You can think of the first tensor as a batch of 29 sentences with 30 words each from a vocabulary of 32 words, and the second tensor as a random embedding matrix.

2. Use `torch.matmul` to matrix multiply the two tensors.

3. What is the difference between `torch.matmul`, `torch.mm`, `torch.bmm`, and `torch.einsum`, and the `@` operator?

4. Use `torch.sum` on the resulting tensor, passing the optional argument of `dim=1` to sum across the 1st dimension. Before you run this, can you predict the size? This step is akin to computing a sentence embedding by adding up all the embeddings of the words in the sentence.

5. Create a new `long` tensor of size `(3, 10)` with random integers between 0 and 30.

6. Use this new long tensor to index into the result tensor from step 2. What is the shape of the output?

7. Use `torch.mean` to average across the penultimate dimension in the tensor from step 6. By analogy with step 4, what does this step do in a language modeling module?

Optimization

Problem 10 [5 points] Write a Python program to compute the value of $\sqrt{2}$ up to 4 decimal places without using the square root or other math functions from any of the libraries. Only include your function. Do not include any code output.
Hint: Notice that the answer is the (positive) root of the equation, $x^2 - 2 = 0$. To find the root, you might want to use "Newton's Method".